oystermapR

Methodological Rationale

Statistical and design decisions — evidence base and alternatives considered

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This document records the rationale behind every statistical method, scoring curve, threshold value, algorithmic choice, and model design decision made in oystermapR. It is intended for scientific peer review, regulatory submission, and to guide future developers maintaining or extending the package. Each section identifies the chosen approach, describes the alternatives considered, and cites the primary evidence base.

Contents

1 Scoring Framework

1.1 Analytic Hierarchy Process (AHP) Weighting

Each environmental variable is assigned a rank by an ecologist, reflecting its relative importance to species survival (rank 1 = most important). The weight assigned to rank r is 1/r, normalised so all weights sum to 1. This is a simplified AHP weight derivation — formally equivalent to the reciprocal rank weighting used in information retrieval.

Decision: Variable weighting scheme

Chosen approach: Rank-based reciprocal weights (1/r, normalised)

Alternatives considered: Equal weights; expert-elicited pairwise comparison matrix (full AHP); machine learning feature importance

- Equal weights ignore well-established ecological hierarchies — temperature and salinity are universally recognised as primary survival determinants for marine bivalves (Helm & Bourne 2004; FAO).
- Full AHP pairwise comparison matrices require expert consensus panels and produce inconsistency ratios that are difficult to validate for five or more criteria. For an open-source tool, transparent rank assignment by the species expert is more reproducible.
- Machine learning feature importance is data-dependent and cannot be applied without labelled training data, which does not exist for most restoration sites.
- Reciprocal rank weighting is a well-accepted proxy for AHP when ordinal ranks are assigned by domain experts (Saaty 1980; Malczewski 1999 GIS-MCDA review).

Saaty (1980): The Analytic Hierarchy Process. McGraw-Hill.

Malczewski (1999): GIS and Multicriteria Decision Analysis. Wiley.

Helm & Bourne (2004): FAO Hatchery Manual — Bivalve Aquaculture.

1.2 Piecewise Linear Scoring Curves

Each numeric variable is scored on [0,1] using a four-zone piecewise linear function: optimal range (score = 1.0), acceptable range (linear decay from 1.0 to 0.0), absolute limit (hard zero). One-sided variants (threshold decay) are used for variables with only an upper limit (e.g. turbidity).

Decision: Shape of scoring curves

Chosen approach: Piecewise linear (trapezoid) with hard limits

Alternatives considered: Gaussian bell curve; sigmoid (logistic); step function; polynomial

- Gaussian curves produce non-zero scores beyond any hard biological limit (e.g. lethal temperature), which is ecologically incorrect.
- Step functions (binary: in range / out of range) discard the ecologically meaningful gradient within the acceptable range — a location at the edge of tolerance is not equivalent to one at the optimum.
- Sigmoid curves are parameterised by inflection point and steepness, which are rarely quantified in bivalve literature. Piecewise linear curves can be parameterised directly from published optimal/acceptable range tables.
- Piecewise linear is the standard formulation in habitat suitability modelling literature and is interpretable without statistical training (Brown 1994; Store & Kangas 2001).

Brown (1994): Predicting viability of fish habitat using habitat suitability index models. Fisheries.

Store & Kangas (2001): Integrating spatial multi-criteria evaluation and expert knowledge in GIS-based habitat suitability modelling. Landscape and Urban Planning.

1.3 Hard Exclusions as Binary Gates

Temperature, salinity, and dissolved oxygen can trigger a hard exclusion — the location receives suitability = 0 regardless of all other variables. These are not scored on the continuous curve.

Decision: Treatment of lethal conditions

Chosen approach: Binary hard exclusion gates for temperature, salinity, dissolved oxygen

Alternatives considered: Low continuous score; weighted penalty; post-hoc masking

- Lethal conditions are not a matter of degree — an animal cannot partially survive a water temperature above its LT50. Applying a low continuous score would still allow a high score from other variables to offset a lethal condition, which is biologically impossible.
- Dissolved oxygen below 4 mg/L causes rapid mortality in O. edulis (Widdows et al. 1989). Turbidity, depth, and substrate cannot compensate for anoxia.
- Hard exclusion is consistent with MaxEnt's hard constraint layer approach and with regulatory frameworks (OSPAR 2009 O. edulis restoration guidelines).

Widdows et al. (1989): Physiological responses of Mytilus edulis to hypoxia. Marine Biology.

OSPAR Commission (2009): Background document on Ostrea edulis and Ostrea edulis beds. OSPAR Biodiversity Series.

1.4 Missing Variable Handling

When a scored variable is absent (NA) for a row, it is excluded from both the numerator and denominator of the weighted mean, and remaining weights are renormalised. The fraction of defined variables with real data is recorded as data_completeness.

Decision: Missing variable imputation strategy

Chosen approach: Weight redistribution with data_completeness flag

Alternatives considered: Mean imputation; zero imputation; neutral (0.5) score; model exclusion

- Zero imputation would penalise locations where a variable simply was not measured, conflating data gaps with poor habitat quality.
- Mean imputation introduces artificial central tendency — a location with unknown turbidity should not be assumed to have average turbidity.
- Neutral score (0.5) arbitrarily inflates scores for data-poor locations.
- Weight redistribution is the principled choice: it scores what is known and flags uncertainty via data_completeness. Users can filter or caution-flag results with completeness < 0.5.

2 Tidal Correction

2.1 Tidal Constituent Selection

The harmonic tidal model uses five constituents: M2 (principal lunar semi-diurnal), S2 (principal solar semi-diurnal), N2 (larger lunar elliptic), K1 (luni-solar diurnal), O1 (principal lunar diurnal). These are combined as $H(t) = Z0 + \Sigma f \cdot A \cdot \cos(\omega t - g + u)$.

Decision: Number and identity of tidal constituents

Chosen approach: 5-constituent model: M2, S2, N2, K1, O1

Alternatives considered: 2-constituent (M2+S2 only); 8-constituent; full 60+ constituent IHO model

- M2 alone accounts for approximately 60-70% of tidal variance in NW European shelf seas. Adding S2 captures the spring-neap cycle (the most ecologically significant tidal variation for intertidal species). N2 adds the elliptic modulation (~4% of M2 amplitude). K1 and O1 are the dominant diurnal components important for sites with mixed tidal regimes (Irish Sea, Solway Firth).
- A 2-constituent model (M2+S2) would miss diurnal inequality in the Irish Sea and produce systematic depth errors up to ~0.4 m at spring tides.
- Full IHO models (60+ constituents) require harmonic analysis of multi-year tide gauge records not available at all 31 reference ports, and the additional precision is below instrument noise for typical ADCP/CTD deployments.
- The 5-constituent set is standard in applied coastal engineering for surveys without long-baseline harmonic records (Pugh 1987; IOC 2016).

Pugh (1987): Tides, Surges and Mean Sea-Level. Wiley.

IOC/SCOR/IAPSO (2016): The international thermodynamic equation of seawater — 2010. Intergovernmental Oceanographic Commission.

2.2 Nodal Corrections (Schureman 1940)

The amplitude and phase of tidal constituents are modulated by the 18.6-year cycle of the ascending lunar node (N). The nodal correction factors f (amplitude) and u (phase) are computed from N at the survey midpoint. N is derived from J2000.0 epoch: $N = 125.04 - 0.052954 \cdot \text{days\_since\_J2000}$.

Decision: Nodal correction implementation

Chosen approach: Schureman (1940) nodal factor formulas evaluated at survey midpoint

Alternatives considered: f=1, u=0 (simplified, no nodal correction); astronomical argument from JPL ephemeris

- Without nodal corrections, M2 amplitude error peaks at ±3.7% near the nodal minimum/maximum (every 9.3 years). For a typical 3 m spring range, this is ±11 cm — comparable to the depth uncertainty budget.
- Full JPL ephemeris computation adds a hard dependency on external data and is unnecessary: the closed-form Schureman equations agree with ephemeris to within 0.01% over century timescales.
- Constituent-specific formulas are used (different f and u for M2/S2, K1, O1, N2) rather than a single mean correction, as recommended by Schureman.

Schureman (1940): Manual of Harmonic Analysis and Prediction of Tides. US Coast and Geodetic Survey Special Publication 98.

2.3 Reference Port Matching and Coverage

Each survey location is matched to its nearest of 31 harmonic reference ports using Euclidean distance in decimal degrees. Surveys beyond 75 km from the nearest port receive a warning, as constituent amplitudes may deviate significantly in offshore or far-field areas.

Decision: Port matching strategy

Chosen approach: Nearest-port Euclidean matching with 75 km warning threshold

Alternatives considered: Inverse distance interpolation between multiple ports; fixed-region lookup table

- Interpolation between ports would require consistent phasing conventions and resolving phase ambiguities across constituent families — a non-trivial problem for a user-facing tool.
- 75 km threshold is conservative: tidal constituent amplitudes vary smoothly over most of the NW European shelf, and the dominant M2 error at 75 km is typically < 5 cm in non-resonant regions.
- The 31-port network covers the UK, Ireland, North Sea, and northern French coast — the primary deployment area for the current species database.

3 Species Tolerance Parameters

3.1 Ostrea edulis — AHP Rank Sources

Variable ranks for O. edulis are based on the published physiological and ecological literature. The ranking reflects lethal vs sublethal vs performance-limiting effects at each life stage.

Variable	Rank	Primary Evidence
Temperature	1	Bayne (1965); Diez et al. (1995): LT50 = 29°C; growth optimum 15-20°C; spawning trigger 16°C.
Salinity	2	Shumway (1996): survival range 20-40 PSU; optimum 28-35 PSU.
Depth	3	OSPAR (2009): O. edulis forms reefs 0-20 m depth; rarely subtidal > 40 m.
Dissolved oxygen	4	Widdows et al. (1989): anaerobic threshold ~4 mg/L; suboptimal < 6 mg/L.
Turbidity	5	Navarro & Iglesias (1993): filtration inhibited > 30 NTU; severe reduction > 50 NTU.
Substrate hardness	6	FAO Hatchery Manual: settles on hard substrate with conspecific shell or biofilm.
Current velocity	7	Wildish & Kristmanson (1997): passive suspension feeders benefit from 0.05-0.5 m/s.
Chlorophyll a	8	Rico-Villa et al. (2009): phytoplankton-driven growth; 1-5 µg/L optimal for growth.
pH	9	Gazeau et al. (2007): ocean acidification effects; < 7.8 reduces calcification.

Bayne (1965): Growth and reproduction of Ostrea edulis. J Marine Biol Assoc UK.

Shumway (1996): Natural Environmental Factors. In: The Eastern Oyster. Maryland Sea Grant.

OSPAR (2009): Background document on Ostrea edulis. OSPAR Biodiversity Series.

3.2 Temperature Seasonality Override

O. edulis has season-dependent temperature tolerances: optimal summer temperature (15-20°C) differs from optimal winter threshold (> 4°C to prevent mortality). When a season column is present, the seasonal variant of the tolerance spec is applied.

Decision: Seasonal temperature handling

Chosen approach: Season-specific scoring curve variant selected from tolerance spec

Alternatives considered: Year-round mean curve; worst-case (winter) curve only; ignore seasonality

- Using a year-round mean curve misrepresents the thermal biology — a temperature of 8°C is near-optimal in winter but suboptimal in summer.
- Worst-case only would systematically undervalue sites with good summer conditions, which is the primary growth season.
- Season detection uses latitude-aware meteorological seasons to handle southern hemisphere deployments correctly.

3.3 Categorical Variable Scoring

Substrate type and benthic community are scored from lookup tables mapping category names to scores. String matching is case-insensitive with partial matching fallback. Unknown categories receive a neutral 0.4 score.

Decision: Unknown category handling

Chosen approach: Neutral score 0.4 (slightly below midpoint)

Alternatives considered: Score of 0 (worst case); score of 0.5 (true neutral); exclude from scoring

- Score of 0 would penalise unmeasured or unidentified substrate types, which is inappropriate — absence of substrate data is not evidence of poor substrate.
- Score of 0.5 (true neutral) was considered but 0.4 was chosen as a slight downward nudge to reflect that uncharacterised substrate is genuinely less desirable for restoration planning than known-good substrate.
- This approach is consistent with the precautionary principle in OSPAR restoration guidelines.

4 Spatial Methods

4.1 Kriging vs IDW for Survey Interpolation

Survey measurements are interpolated to a regular grid using ordinary kriging (when ≥ 50 observations are available) or IDW as a fallback. The decision to prefer kriging is based on its statistical optimality properties.

Property	Ordinary Kriging	IDW (p=2)
Statistical basis	Best Linear Unbiased Estimator (BLUE); minimises mean squared prediction error under stationarity assumption	No statistical optimality; empirical distance-decay rule
Uncertainty output	Returns kriging variance (prediction uncertainty) at every cell	No uncertainty estimate
Minimum data requirement	≥50 observations for reliable variogram fitting	As few as 3 observations
Computational cost	Moderate (variogram fitting $O(n \log n)$; kriging $O(n^2)$ per grid cell)	Fast $O(n)$ per grid cell
Artefacts	Smooth, honours spatial autocorrelation structure	Bull's-eye artefacts around isolated points
Variogram model	Auto-fitted Matérn (κ=1.5), fallback spherical; both physically interpretable for marine environmental data	N/A
External dependency	gstat + sp R packages	None (implemented internally)

The Matérn covariance model (κ=1.5, equivalent to the Whittle-Matérn model) is preferred over spherical for environmental data because it is infinitely differentiable at all lags (ensuring smooth prediction surfaces), and its smoothness parameter κ has a physical interpretation related to the differentiability of the underlying process (Stein 1999). The spherical model is used as a fallback when Matérn fitting fails (numerically unstable at very small sample sizes).

Cressie (1993): Statistics for Spatial Data. Wiley.

Stein (1999): Interpolation of Spatial Data. Springer.

Shepard (1968): A two-dimensional interpolation function for irregularly-spaced data. Proc. ACM National Conference.

4.2 Gaussian Kernel Spatial Smoothing

The smooth_suitability() function replaces each cell's score with a distance-weighted mean of all cells within the search radius, using Gaussian weights $w = \exp(-d^2 / 2\sigma^2)$.

Decision: Spatial smoothing kernel

Chosen approach: Gaussian kernel with configurable bandwidth σ

Alternatives considered: Simple moving average (equal weights); median filter; no smoothing

- Equal-weight moving average over a square window produces anisotropic smoothing — cells at the corner of the window receive the same weight as cells immediately adjacent, which is geometrically incorrect.
- Median filter is robust to outliers but non-linear — it does not preserve the weighted mean properties needed for score interpretation.
- Gaussian weighting is isotropic, continuously differentiable, and produces smooth output surfaces consistent with the assumption that environmental conditions vary smoothly over survey scales.
- Excluded cells are not allowed to contribute to their neighbours' smoothed scores by default — this preserves hard barriers (anoxic basins, freshwater outfalls) rather than blurring them out.

4.3 Connectivity Algorithm (Union-Find)

Habitat patches are identified using a union-find (disjoint set) algorithm. Two suitable cells are in the same patch if they are within gap_m metres of each other. The default 500 m gap threshold corresponds to approximately one tidal excursion at a moderate current speed.

Decision: Graph connectivity algorithm

Chosen approach: Union-find with path compression; no external graph package

Alternatives considered: Breadth-first search (BFS); DBSCAN clustering; igraph-based graph analysis

- BFS on a distance matrix would require $O(n^2)$ memory for the full adjacency matrix, which is prohibitive for surveys with > 1000 cells.
- DBSCAN (density-based clustering) uses the same distance threshold concept but is designed for point cloud clustering rather than habitat connectivity, and its ε and min_pts parameters do not map cleanly onto ecological connectivity.
- igraph provides richer graph analysis but adds a large binary dependency inappropriate for a lightweight R package.
- Union-find with path compression achieves $O(n\,\alpha(n))$ amortised complexity (effectively linear) with no external dependencies.
- 500 m gap default is ecologically grounded: O. edulis veliger larvae have a planktonic phase of 10-14 days and are transported ~100-500 m per tidal cycle in typical shelf sea currents (Bayne 1965; Laing et al. 1997).

Laing et al. (1997): Effects of temperature and food on larval settlement and metamorphosis of the European flat oyster, Ostrea edulis. J Shellfish Res.

5 Uncertainty Quantification

5.1 Bootstrap Design

Uncertainty bounds are produced by add_suitability_ci(), which re-scores each location n_boot times after adding independent Gaussian noise to every numeric variable. The 5th and 95th percentiles form the 90% confidence interval.

Decision: Uncertainty quantification method

Chosen approach: Parametric bootstrap with Gaussian measurement noise

Alternatives considered: Analytical error propagation; Monte Carlo with non-parametric resampling; no uncertainty

- Analytical error propagation through piecewise linear functions is tractable but produces symmetric intervals that underestimate uncertainty near breakpoints (where the derivative is discontinuous).
- Non-parametric resampling (case bootstrap) resamples entire rows, producing spatial uncertainty rather than measurement uncertainty. This conflates spatial variability with instrument error.
- Gaussian noise is appropriate for CTD and ADCP sensors, which have well-characterised instrument uncertainty specifications (e.g. RBR CTD: ±0.002°C temperature, ±0.003 PSU salinity).
- 90% interval (5th-95th percentile) is chosen over 95% (2.5th-97.5th) because instrument errors in oceanographic deployments are rarely Gaussian tails — a 90% interval is more conservative without overstating precision.

5.2 Iteration Count

Decision: Default bootstrap iteration count

Chosen approach: n_boot = 200

Alternatives considered: n=50 (fast preview); n=500 (publication); n=1000 (high precision)

- Bootstrap percentile estimates stabilise when the Monte Carlo standard error of the 5th/95th percentile is < 0.005 suitability units. At n=200, MCSE ≈ 0.003 for typical suitability distributions.
- n=50 produces visually acceptable CIs for exploratory work but is too variable for regulatory submission.
- n=500 is recommended for final reporting and matches the default in ecological SDM uncertainty literature (e.g. Thuiller et al. 2019).
- Users can override: n_boot=50 for quick runs during model development, n_boot=500 for peer-reviewed outputs.

Thuiller et al. (2019): Uncertainty in ensembles of global biodiversity scenarios. Nature Communications.

6 Validation and Model Diagnostics

6.1 ROC / AUC

The receiver operating characteristic (ROC) curve plots sensitivity (true positive rate) against 1-specificity (false positive rate) across all possible suitability thresholds. The area under the curve (AUC) is computed by the trapezoidal rule. AUC is threshold-independent and scale-invariant — it measures the probability that a random presence site scores higher than a random absence site.

Decision: Discrimination metric

Chosen approach: AUC (trapezoidal ROC integral)

Alternatives considered: Accuracy at fixed threshold; Cohen's kappa; Pearson correlation; log-likelihood

- Accuracy at a fixed threshold is sensitive to the choice of threshold and to prevalence imbalance (many absences, few presences — common in restoration surveys).
- AUC is prevalence-independent and threshold-free, making it comparable across surveys with different presence/absence ratios.
- AUC is the standard metric in species distribution modelling literature (Fielding & Bell 1997; Elith et al. 2006; Lobo et al. 2008).

Fielding & Bell (1997): A review of methods for the assessment of prediction errors in conservation presence/absence models. Environmental Conservation.

Elith et al. (2006): Novel methods improve prediction of species distributions from occurrence data. Ecography.

6.2 Youden's J for Optimal Threshold Selection

Decision: Threshold selection criterion

Chosen approach: Youden's J = sensitivity + specificity – 1 (maximised)

Alternatives considered: F1 maximisation; cost-minimisation; fixed threshold (e.g. 0.5); ROC distance to (0,1)

- F1 maximisation is asymmetric — it weights precision and recall equally but ignores specificity, which is important when absence records are meaningful (e.g. dredge survey zeroes from known beds).
- A fixed threshold of 0.5 is arbitrary and ignores the actual score distribution.
- Youden's J is the threshold at the point on the ROC curve farthest from the diagonal (random classifier), geometrically equivalent to maximising the vertical distance to the random line. It treats sensitivity and specificity symmetrically, consistent with equal costs for false positives and false negatives in restoration site selection.
- Youden's J is the most widely recommended threshold criterion in the absence of cost information (Fluss et al. 2005).

Fluss et al. (2005): Estimation of the Youden Index and its associated cutoff point. Biometrical Journal.

6.3 True Skill Statistic (TSS)

TSS = sensitivity + specificity – 1. Evaluated at the optimal Youden threshold. TSS ranges from –1 (perfectly wrong) through 0 (random) to 1 (perfect). Unlike kappa, TSS is not affected by prevalence.

Decision: Single-number skill metric at optimal threshold

Chosen approach: TSS (True Skill Statistic)

Alternatives considered: Cohen's kappa; Heidke Skill Score; accuracy

- Kappa is sensitive to prevalence (Allouche et al. 2006) — its value changes with the ratio of presences to absences even for the same underlying skill, making it unsuitable for comparing models across surveys with different presence/absence ratios.
- TSS is prevalence-independent and directly interpretable: TSS > 0.4 is the commonly cited threshold for useful predictive skill (Landis & Koch 1977 kappa re-interpretation applied to TSS by Allouche et al. 2006).

Allouche et al. (2006): Assessing the accuracy of species distribution models: prevalence, kappa and the true skill statistic (TSS). J Applied Ecology.

6.4 Brier Score

Brier score = mean((predicted_suitability – presence)^2). Treats the suitability score as a probability estimate. 0 = perfect; 0.25 = random (for 50/50 prevalence); > 0.25 = worse than random.

Decision: Calibration / probability score

Chosen approach: Brier score

Alternatives considered: Log-loss; reliability diagram; calibration slope

- Log-loss is undefined when predicted probability = 0 or 1, which occurs for excluded cells (suitability = 0). Brier score handles these gracefully.
- Brier score is a proper scoring rule (rewarding calibration), scale is interpretable (random ≈ 0.25 for balanced data), and has a long track record in meteorological and ecological forecast evaluation (Brier 1950; Wilks 2011).

Brier (1950): Verification of forecasts expressed in terms of probability. Monthly Weather Review.

6.5 No External Package Dependency for ROC

ROC, AUC, sensitivity, specificity, and all validation metrics are computed from first principles in base R. This is a deliberate design decision.

Decision: Validation implementation strategy

Chosen approach: From-scratch base R implementation

Alternatives considered: pROC package; ROCR package; MLmetrics

- pROC adds a binary package dependency with its own C-level code, version management overhead, and potential CRAN policy implications.
- A from-scratch implementation with trapezoidal AUC, Youden threshold, and manual confusion matrix is < 80 lines of R and is fully auditable — important for regulatory submissions where methodology must be transparent.
- The trapezoidal AUC is mathematically equivalent to pROC's default method for discrete score distributions (DeLong et al. 1988).

DeLong et al. (1988): Comparing the areas under two or more correlated receiver operating characteristic curves. Biometrics.

6.6 Spatial Block Cross-Validation (Roberts et al. 2017)

Standard k-fold cross-validation assigns records to folds randomly, so nearby train and test records share similar environments — the model 'leaks' through spatial autocorrelation and AUC is inflated. Spatial block CV partitions records into contiguous geographic blocks and uses leave-one-block-out evaluation, giving honest out-of-sample performance.

Decision: Cross-validation strategy

Chosen approach: Spatial block CV (leave-one-block-out, k=5 default blocks)

Alternatives considered: Random k-fold CV; jackknife; temporal split

- Roberts et al. (2017) demonstrated that random CV overestimates transferability by a mean of 0.11 AUC units for species distribution models with spatial autocorrelation ranges > 50 km. Oyster survey data from acoustic survey vessels typically shows autocorrelation ranges of 1–20 km.
- The spatial autocorrelation range is estimated from Moran's-I style pairwise distance binning of suitability scores in base R, without requiring the gstat or spdep packages.
- A spatial bias warning fires automatically when the estimated autocorrelation range exceeds 0.05° (approx. 5 km).

Roberts D.R. et al. (2017): Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40:913-929.

Hijmans R.J. (2012): Cross-validation of species distribution models: removing spatial sorting bias. Ecology 93:679-688.

6.7 Permutation Variable Importance

For each scored variable, values are randomly shuffled across all prediction cells N times (default 50) and AUC is recomputed. Importance = baseline_AUC − mean(permuted_AUC). This is the Breiman (2001) permutation importance approach applied to the AHP scoring function.

Importance drop	Interpretation

> 0.10 (10% AUC drop)	High importance — variable is load-bearing for discrimination.
0.02 – 0.10	Moderate importance.
< 0.02	Low importance — consider omitting to reduce data collection cost.

Breiman L. (2001): Random Forests. Machine Learning 45:5-32.

Zurell D. et al. (2020): A standard protocol for reporting species distribution models (ODMAP). Ecography 43:1261-1277.

6.8 Partial Dependence / Sensitivity Analysis

sensitivity_analysis() computes the marginal response curve for a single variable: suitability is predicted at a grid of values while all other variables are held at their observed medians. This verifies that the AHP scoring function produces ecologically reasonable response shapes and is required by the ODMAP reporting protocol (Zurell et al. 2020) for published species distribution models.

7 Multi-species and Competition

7.1 M. gigas Competition Penalty on O. edulis

When both M. gigas and O. edulis are scored, a post-hoc multiplier is applied to O. edulis suitability where M. gigas scores High (multiply by 0.80) or Moderate (multiply by 0.92).

Decision: Competition modelling approach

Chosen approach: Post-hoc suitability multiplier based on M. gigas suitability class

Alternatives considered: Mechanistic competition model; Lotka-Volterra interaction term; no adjustment

- Mechanistic competition models require parameterised interaction coefficients (α in Lotka-Volterra) that have not been quantified for this species pair in field conditions at landscape scale.
- The 0.80 penalty for High M. gigas sites reflects the well-documented competitive exclusion of O. edulis in mixed beds — M. gigas grows faster, filters more efficiently, and produces denser beds that physically displace flat oysters (Wrange et al. 2010; Zwerschke et al. 2018).
- The 0.92 penalty for Moderate M. gigas sites is a conservative intermediate, acknowledging that moderate M. gigas density allows some O. edulis persistence (Zwerschke et al. 2020 mesocosm data).
- The original O. edulis score is preserved in suit_ostrea_edulis so users can inspect both the raw ecological suitability and the competition-adjusted estimate.

Wrange et al. (2010): Foul or fair? The expansion of the Pacific oyster Crassostrea gigas in Scandinavia. Biological Invasions.

Zwerschke et al. (2018): Co-occurrence of native Ostrea edulis and non-native Crassostrea gigas revealed by monitoring. J Sea Research.

Zwerschke et al. (2020): Competition between a native and non-native benthic species increases with elevated CO2. ICES J Marine Science.

7.2 Spat Settlement Tolerance Differences

The settlement scoring module uses a separate, stricter tolerance spec. Key differences from the adult spec are justified as follows:

Parameter	Adult spec	Settlement spec	Rationale
Turbidity hard limit	50 NTU	20 NTU	Silt deposition smothers post-set spat (Robert et al. 2017).
Current velocity minimum	None	0.05 m/s	Still water prevents larval delivery to attachment substrate (Bayne 1965).
Temperature optimum (O. edulis)	15-20°C	16-22°C	Metamorphosis requires warmer temperatures than adult performance optimum (Laing et al. 1997).
Substrate hard limit	No hard limit	Very Soft = 0	Veliger larvae cannot attach to fluid mud; requirement for biofilm on hard surface is absolute (Helm & Bourne 2004).

Robert et al. (2017): Combined effects of temperature and turbidity on bivalve larval settlement. Marine Environmental Research.

Helm & Bourne (2004): FAO Hatchery Manual — Bivalve Aquaculture. FAO Fisheries Technical Paper 471.

7.3 Seasonal Composite Method Selection

Decision: Seasonal composite aggregation method

Chosen approach: min (default), mean, or prob — user selectable

Alternatives considered: Geometric mean; weighted by season length; harmonic mean

- min is ecologically the most defensible for restoration site selection: oysters cannot survive an unsuitable season regardless of how suitable other seasons are. A site that becomes anoxic in summer is not a restoration candidate even if spring conditions are excellent.
- mean is appropriate for monitoring trend summaries where partial coverage across seasons is acceptable.
- prob (fraction of seasons above threshold) is appropriate when surveys are irregularly distributed — it avoids punishing a site for one anomalous survey in an otherwise suitable record.
- Geometric mean was considered but discarded because it approaches zero if any season has near-zero suitability, behaving similarly to min but less interpretably.

8 Environmental Risk Layers

8.1 Bonamia ostreae Risk Curve Parameterisation

The Bonamia risk curve uses temperature as the primary driver. Key thresholds are: cold safe ≤ 8°C (near-zero parasite proliferation), onset 10°C, peak transmission 15-20°C, slight reduction above 22°C.

Threshold	Value	Source
Cold safe (no proliferation)	≤ 8°C	Culloty & Mulcahy (2007): proliferation rate ~0 at 5-8°C in haemocyte cultures.
Transmission onset	10°C	Arzul et al. (2009): Bonamia detected in oysters exposed > 10°C.
Peak transmission window	15-20°C	ICES WGOS (2022): outbreak years consistently coincide with SST > 15°C in June-August.
Salinity modulation onset	25 PSU	Cochennec-Laureau et al. (2003): lower salinity reduces Bonamia transmission probability.

Culloty & Mulcahy (2007): Bonamia ostreae in Ostrea edulis: a review. Hydrobiologia.

Arzul et al. (2009): Contribution to the understanding of the cycle of the haplosporidian Bonamia ostreae. J Invertebrate Pathology.

ICES WGOS (2022): Report of the Working Group on Oyster and Shellfish Ecology. ICES CM 2022.

8.2 Climate Scenario Alignment (UKCP18)

Built-in scenarios use SST and salinity deltas aligned with UKCP18 marine projections for NW European shelf seas at the 2050s horizon.

Scenario	∆SST	∆Salinity	UKCP18 alignment
rcp26 (SSP1)	+0.5°C	0.0 PSU	UKCP18 10th percentile ensemble, low-emission pathway
rcp45 (SSP2)	+1.2°C	–0.2 PSU	UKCP18 50th percentile (median) ensemble
rcp85 (SSP5)	+2.5°C	–0.5 PSU	UKCP18 90th percentile, high-emission worst case

Salinity decrease under warming reflects increased freshwater input from enhanced North Atlantic precipitation and ice melt (Lemasson et al. 2017). Custom deltas can be supplied to override built-ins with any downscaled regional projection.

Met Office (2018): UKCP18 Marine Projections. Available at: ukclimateprojections.metoffice.gov.uk

Lemasson et al. (2017): Linking the biological impacts of ocean acidification on oysters to changes in carbonate chemistry. J Experimental Marine Biology and Ecology.

8.3 QC Outlier Detection Threshold (Tukey Outer Fence)

Decision: Outlier detection threshold

Chosen approach: Tukey outer fence: k=3 × IQR from median

Alternatives considered: k=1.5 (inner fence, standard boxplot); z-score > 3; Grubbs test; GESD

- k=1.5 (the standard boxplot whisker) flags ~0.7% of values from a normal distribution as outliers, which is too aggressive for typical ADCP and CTD data where real environmental gradients can look like statistical outliers.
- k=3 (outer fence) flags only values that are extreme outliers — roughly the top/bottom 0.02% of a normal distribution — appropriate for sensor spike detection without excessive false positives.
- Z-score outlier detection assumes normality, which is not appropriate for turbidity or chlorophyll (typically right-skewed distributions).
- k is configurable by the user (reduce to 1.5 for high-precision CTD data where spikes are small in absolute magnitude).

Tukey (1977): Exploratory Data Analysis. Addison-Wesley.

8.4 Wave Exposure — JONSWAP Fetch-Growth Model

Significant wave height Hs at each survey location is estimated from effective fetch F (km) and wind speed U (m/s) using the JONSWAP fetch-growth empirical formula (Hasselmann et al. 1973):

$Hs = 0.0248 \cdot (U^2/g) \cdot (gF/U^2)$■■■² with depth cap: $Hs\_max = 0.6 \cdot depth\_m$ (depth-limited breaking).

Parameter	Value / Source
g (gravitational acceleration)	9.81 m/s²
Fetch scaling exponent	0.42 (Hasselmann et al. 1973 JONSWAP dataset)
Default wind speed	12 m/s (representative UK coastal mean; user-overridable)
Depth-limited cap	0.6 × depth: Munk-Traylor (1947) modified breaking criterion
Wave score breakpoints	Hs < 0.5 m = Sheltered; < 1.5 m = Moderate; < 2.5 m = Exposed; ≥ 2.5 m = Severe

Hasselmann K. et al. (1973): Measurements of wind-wave growth and swell decay (JONSWAP). Dtsch Hydrogr Z Suppl A8.

8.5 Sediment Stability — Shields Parameter Analysis

Whether oyster spat and reef substrate remain in place depends on bed shear stress relative to the threshold for sediment entrainment. The Shields dimensionless parameter θ is the standard formulation:

$\theta = \tau\_b / ((R\_s - R\_w) \cdot g \cdot d50)$ where $\tau\_b = R\_w \cdot C\_f \cdot U\_combined^2$; $U\_combined = \sqrt{U\_current^2 + U\_wave\_orbital^2}$.

Decision	Rationale
Critical Shields θ_cr: van Rijn / Soulsby (1997) formula	Accounts for particle Reynolds number $D^* = d50 \cdot ((R \cdot g)/\nu^2)^{1/3}$; more accurate than fixed $\theta\_cr = 0.047$ for fine/coarse sediment mix.
Substrate → d50 lookup table	Very soft mud = 0.010 mm; soft mud = 0.080 mm; sand = 0.300 mm; mixed = 3.000 mm; hard/gravel = 25 mm. Derived from Wentworth scale modal values.
Wave orbital velocity from dispersion relation	Iterative shallow-water dispersion k (10 steps); $U\_w = \pi H/(T \cdot \sinh(kd))$. Avoids incorrect deep-water-only formula below 30 m.
Mobility ratio gate	$\theta/\theta\_cr < 0.5$ = Stable; < 1.0 = Marginally stable; < 2.0 = Mobile; ≥ 5.0 = Highly mobile (score = 0).

van Rijn L.C. (1984): Sediment transport, Part I. J Hydraulic Engineering 110:1431-1456.

Soulsby R.L. (1997): Dynamics of Marine Sands. Thomas Telford, London.

8.6 HAB Risk — Closure Frequency and Toxin Weighting

Harmful Algal Bloom risk is scored on a [0,1] scale combining: (1) historical closure frequency from ICES HAB / CEFAS biotoxin records; (2) dissolved inorganic nitrogen (eutrophication driver); (3) thermal stratification index (surface-bottom $\Delta T > 2°C$).

Toxin class	Severity weight	Rationale
PSP (Alexandrium)	1.00	Complete closure; no depuration threshold. Immediate human health risk.
AZP (Azaspiracid)	0.90	Persistent; low depuration rates in flat oysters (EFSA 2008).
ASP (Pseudo-nitzschia)	0.90	Domoic acid; oyster depuration possible but slow.
DSP (Dinophysis)	0.80	OA toxins; regular UK closures; depuration feasible.
Unknown/other	0.60	Conservative default when genus not recorded.

Normalisation: 30 closure days/year maps to raw score 1.0 — this is the empirical threshold above which a site is typically deemed operationally unviable for aquaculture.

ICES WGHABD (2022): Working Group on Harmful Algal Bloom Dynamics. CM 2022.

EFSA Panel on Contaminants (2008): Opinion on azaspiracids. EFSA Journal 723:1-52.

8.7 Anthropogenic Disturbance — ICES VMS Swept-Area Ratio

Bottom trawling intensity is quantified using the ICES VMS Swept-Area Ratio (SAR): total seabed area swept per unit area per year. SAR > 0.5/yr triggers the gear deployment gate (disturbance_gear_gate = TRUE) — above this threshold, physical gear deployment is not viable without regulatory exclusion (Hiddink et al. 2017).

SAR threshold	Score	Class
≤ 0	0.00	Negligible
0.1	0.20	Low
0.5	0.60	Moderate
1.0	0.85	High
≥ 3.0	1.00	Active (gear gate triggered)

Gear impact multipliers: dredge ×1.4, beam ×1.2, otter ×1.0, seine ×0.8. These reflect relative seabed contact area per tow (Eigaard et al. 2017). Dredge gear causes the greatest surface disturbance and is the dominant pressure on historic flat oyster habitats in the British Isles.

Eigaard O.R. et al. (2017): The footprint of bottom trawling in European waters. ICES J Mar Sci 74:1700-1710.

Hiddink J.G. et al. (2017): Global analysis of depletion and recovery of seabed biota. PNAS 114:8301-8306.

8.8 Predation Risk — Species-Specific Predator Weighting

Predation risk integrates recorded predator density (from field survey or EMODnet Biology), depth-based proxy (shallow ≤ 5 m = high), and substrate rugosity factor (hard substrate provides refuge).

Predator taxon	O. edulis weight	M. gigas weight	Rationale
Asterias rubens (common starfish)	1.00	0.60	Primary flat oyster predator; causes mass mortality events.
Urosalpinx cinerea (American oyster drill)	0.90	0.90	Invasive; highly efficient oyster predator in UK/NL populations.
Carcinus maenas (shore crab)	0.80	0.70	Effective on juvenile oysters < 30 mm; reduces on rocky substrate.
Ocenebra erinaceus (sting winkle)	0.70	0.50	Native drill; lower density but persistent pressure.

Smaal A.C. et al. (2019): Goods and Services of Marine Bivalves. Springer.

9 Shellfish Water Quality Classification

EC Regulation 854/2004 (retained in UK law post-Brexit) classifies shellfish harvesting areas into four categories with progressively more restrictive post-harvest treatment requirements.

Class	Status	Economic penalty	Condition
A	Direct harvest permitted	1.00 (none)	< 230 E. coli / 100 g flesh & liquid
B	Requires depuration centre	0.80	< 4,600 E. coli in 90% of samples
C	Relay in open sea ≥ 2 months	0.60	< 46,000 E. coli in 90% of samples
Prohibited	No harvesting permitted	0.00	All others
Unclassified	No data available	0.70	Precautionary default (conservative)

Decision: Unclassified penalty: 0.70 vs 0.50

Chosen approach: Precautionary 0.70 (conservative but not prohibitive)

Alternatives considered: 0.50 (treat as unknown); 0.00 (treat as prohibited)

- An unclassified site is typically unclassified because no monitoring programme exists, not because it is known to be contaminated. 0.70 reflects residual viability while discouraging deployment without prior classification — incentivising applicants to obtain classification before committing infrastructure investment.
- Setting 0.00 would incorrectly eliminate potentially excellent sites from new areas where monitoring has never been carried out.

European Commission (2004): Regulation 854/2004 — hygiene of food of animal origin. Off J Eur Union L226:83-127.

Food Standards Agency (2024): Shellfish classification — food.gov.uk.

10 Bayesian Tolerance Parameter Updating

Species distribution model parameters are inherently uncertain. The expert-elicited tolerance ranges (optimal temperature windows, salinity limits, etc.) are well-grounded in published biology but represent prior beliefs, not posterior estimates. As field presence/absence records accumulate from new surveys, Bayes' theorem provides a principled framework for updating these parameters while preserving scientific priors:

$p(\theta \mid \text{data}) \propto p(\text{data} \mid \theta) \cdot p(\theta)$

where θ is the vector of tolerance parameters (optimal_min, optimal_max per variable), and the likelihood $p(\text{data}|\theta)$ links predicted suitability to observed presence/absence via a Bernoulli logistic model.

10.1 Likelihood Model

$Y_i \sim \text{Bernoulli}(p_i)$, $\text{logit}(p_i) = \alpha + \beta \cdot S_i(\theta)$

$S_i(\theta)$ is the AHP-weighted suitability score for observation i computed under parameter vector θ. α is an intercept (log-odds baseline) and β is a gain term (how sharply suitability predicts presence). Both α and β are estimated jointly with the tolerance parameters.

10.2 MAP + Laplace Approximation (Default)

Decision: Default method: MAP + Laplace approximation

Chosen approach: Find $\theta^* = \text{argmax} \log p(\theta|\text{data})$ via optim() L-BFGS-B; approximate posterior as $MVN(\theta^*, -H^{-1})$ at MAP

Alternatives considered: Full MCMC (Stan/JAGS); Variational inference; Score matching

- Laplace approximation is the same method used internally by brms, INLA, and glmer. It is exact when the posterior is unimodal and Gaussian, and slightly over-confident when skewed — which is typical when datasets are small.
- L-BFGS-B (limited-memory BFGS with box constraints) is used because tolerance parameters have natural box constraints (optimal_min must remain below optimal_max; both must stay within the biological range of the species). No external dependencies required — base R only.
- Full MCMC (Stan/JAGS) requires external software installation and hours of computation for large parameter sets. This is inappropriate as a default for field practitioners. The optional RWMH MCMC sampler is provided for publication-grade credible intervals.

10.3 Sequential Updating

Each call to update_species_tolerances() stores the MAP estimate as the new prior mean. Subsequent calls with new field data automatically treat the previous posterior as the current prior — online Bayesian learning. This is valid when observations are exchangeable (each survey is an independent draw from the same environmental process) and correctly handles multi-season data collection without batch reprocessing.

10.4 Goodness-of-Fit Diagnostic: McFadden Pseudo-R^2

McFadden R^2	Interpretation
< 0.10	Poor fit — parameters barely improve on intercept-only model. Collect more records.
0.10 – 0.20	Acceptable — parameters have some predictive signal.
0.20 – 0.40	Good — comparable to $R^2 > 0.5$ in OLS regression for binary outcomes.
> 0.40	Excellent for SDMs — strong parameter-data congruence.

Gelman A. et al. (2013): Bayesian Data Analysis, 3rd ed. CRC Press.

Rue H. et al. (2009): Approximate Bayesian inference using INLA. JRSS-B 71:319-392.

11 Live Data Integration

All live data fetches are disabled by default (fetch_live = FALSE). Credentials are stored in R options() via oystermapR_live_config() and are never written to disk automatically. Each live source degrades gracefully on failure: the function emits a cli_warn() and continues with either manual data or a conservative default score.

Source	API / Protocol	Data provided	Function
CMEMS (Copernicus Marine)	REST + OGC WCS	Temperature, salinity, current, chlorophyll (satellite + reanalysis)	fetch_live_environmental_data()
ICES HAB Database	GeoServer WFS	Harmful algal bloom event records by species and toxin class	score_hab_risk(fetch_live=TRUE)
ICES VMS (Swept-Area)	GeoServer WFS / OGC	Bottom trawling SAR at c-square resolution (~0.05°)	score_anthropogenic_disturbance(fetch_live=TRUE)
EMODnet Biology	REST API	Species occurrence records (predator density, presence)	score_predation_risk(fetch_live=TRUE)
FSA / DAERA	Open data REST	Shellfish harvesting area classification (England/Wales/NI)	add_shellfish_classification(fetch_live=TRUE)

Decision: Offline-first architecture

Chosen approach: All functions work without internet; live data enriches but never blocks

Alternatives considered: Require-API architecture (fails without credentials)

- The primary users of oystermapR include field ecologists and conservation practitioners working in remote coastal locations with limited or no internet connectivity. Requiring live API calls would make the package unusable for this user group.
- All scored variables accept manually downloaded CSV or dataframe inputs as a first-priority alternative to live fetches, preserving full functionality in air-gapped or restricted-network environments.

Copernicus Marine Service (2024): CMEMS Ocean Data. Available at: marine.copernicus.eu

ICES (2024): ICES Data Portals — VMS and HAB databases. Available at: ices.dk/data

9 Planning and Economics Module

9.1 Gear Feasibility Thresholds

Depth, current velocity, and substrate constraints for each gear type are sourced from published aquaculture engineering guidelines.

Gear type	Depth range (CD)	Current range	Substrate	Source
Longline	4-25 m	0.05-0.60 m/s	Any	FAO (2004); Falconer et al. (2013)
Bottom culture	1-12 m	0.02-0.40 m/s	Hard/mixed	SAMS/Marine Scotland (2022)
Intertidal rack	–2-2 m (MHWS)	0-0.80 m/s	Hard/mixed/sand	Crown Estate (2012)
Raft/pontoon	5-20 m	0.02-0.30 m/s	Any	FAO (2004); Hsiao et al. (2019)
Restoration reef	0-20 m	0.02-0.80 m/s	Hard/mixed	OSPAR (2009); LIFE Re-Seed (2022)

FAO (2004): Oyster Aquaculture: A Practical Manual. FAO Technical Paper No. 471.

Falconer et al. (2013): Modelling support for siting of shellfish farms. Aquaculture Environment Interactions.

LIFE Re-Seed Project (2022): Native oyster restoration guidelines for European coastal waters.

9.2 Economic Viability Index Weights

The viability index combines four components. Weights differ between restoration and aquaculture target modes.

Component	Restoration weight	Aquaculture weight	Rationale
Ecological suitability	45%	30%	Restoration primary objective is long-term species establishment; aquaculture can compensate with management intensity.
Gear feasibility	15%	30%	Restoration reefs have wide gear tolerance; aquaculture requires specific equipment deployment.
Patch area (log-scaled)	25%	20%	Large connected patches are the primary restoration conservation target (connectivity); for aquaculture, smaller intensively managed sites are viable.
Harbour access	15%	20%	Operational cost of vessel access is a secondary consideration for restoration but primary for commercial viability.

Patch area is log-scaled (log10 of km² + 0.001) so that a 10-fold increase in area produces a proportional increase in score — consistent with the species-area relationship underpinning connectivity theory (MacArthur & Wilson 1967). Harbour access is scored on a power-law decay (1 – (d/50)^0.7) to reflect diminishing returns of proximity beyond ~5 km.

MacArthur & Wilson (1967): The Theory of Island Biogeography. Princeton University Press.

9.3 Separation of Planning from Core Ecological Output

The planning module functions (assess_gear_feasibility, score_economic_viability) are implemented as separate post-processing steps rather than integrated into predict_oyster() output. This is a deliberate design decision.

Decision: Integration of planning metrics

Chosen approach: Separate optional functions, not embedded in predict_oyster()

Alternatives considered: Built-in flag in predict_oyster(enable_planning=TRUE); separate R package; hidden columns

- The primary use case for oystermapR is native oyster restoration, where economic viability metrics are irrelevant and would clutter output. Separation keeps the core output clean for conservation users.
- Separate functions allow independent updating of gear specs and viability weights without touching the validated ecological model.
- A hidden enable_planning flag in predict_oyster() was considered but rejected because it would make the function interface ambiguous — it is clearer that planning metrics are a post-processing layer, not part of the ecological suitability model.

13 Larval Dispersal and Population Connectivity

Larval dispersal determines whether a restored or newly seeded oyster bed can self-sustain and whether it can contribute larvae to adjacent habitat. oystermapR v0.4.0 implements a hybrid mechanistic-proxy framework via score_larval_connectivity(), which offers two complementary routes: a union-find spatial clustering algorithm with a Gaussian decay kernel (the primary route), and integration of an external connectivity matrix from particle tracking models such as OpenDrift or FVCOM (the optional advanced route). Both routes can be combined.

13.1 Pelagic Larval Duration and Species-Specific Dispersal

The maximum plausible dispersal distance is calculated as dispersal_km = PLD_days x tidal_excursion_km. Pelagic larval duration (PLD) is the primary determinant of dispersal range. Each of the five supported species has a distinct PLD and larval feeding strategy:

Species	PLD (days)	Larval type	Typical dispersal_km (tidal_excursion = 5 km/day)
Ostrea edulis	4	Lecithotrophic	20 km
Magallana gigas	21	Planktotrophic	105 km
Crassostrea angulata	14	Planktotrophic	70 km
Ostrea stentina	5	Lecithotrophic	25 km
Ostrea lurida	7	Lecithotrophic	35 km

Decision: PLD source values

Chosen approach: Species-specific literature values per .pld_params internal table

Alternatives considered: Single fixed value for all species; user-supplied PLD only; dynamic temperature-dependent PLD

- O. edulis larvae are lecithotrophic (yolk-fed) with PLD ~4 days. M. gigas larvae are planktotrophic with PLD ~21 days at ambient temperatures (Bayne 2017; Cowen & Sponaugle 2009). Using a single PLD for all species would erroneously equate O. edulis (local recruitment) with M. gigas (regional connectivity).
- Temperature-dependent PLD models exist (O'Connor et al. 2007) but require in-situ temperature time-series and substantially increase model complexity. The fixed-PLD approach is consistent with the AHP framework philosophy: biologically grounded default parameters with optional user override.
- Users can override with dispersal_km directly or via pld_days * tidal_excursion_km to capture site-specific current regimes.

Bayne B.L. (2017): Biology of Oysters. Academic Press.

Cowen R.K. & Sponaugle S. (2009): Larval dispersal and marine population connectivity. Ann Rev Mar Sci 1:443-466.

O'Connor M.I. et al. (2007): Temperature control of larval dispersal and implications for marine ecology. PNAS 104:1266-1271.

13.2 Union-Find Spatial Clustering with Gaussian Decay Kernel

Route 1 identifies suitable source patches using a minimum suitability threshold (default min_suitability = 0.40), then groups patches within dispersal_km of each other into connected clusters via union-find (disjoint-set union with path compression). For each target location, the connectivity score reflects three components:

- **Source quality score** (weight 0.50): Gaussian-kernel-weighted mean suitability of all source sites within dispersal_km. Kernel: w(d) = exp(-(d/sigma)^2) where sigma = dispersal_km / 2. Favours sources that are close and of high quality.
- **Cluster size score** (weight 0.30): log-normalised cluster size, capped at 10 sites. Larger connected networks of suitable habitat confer greater long-term recruitment resilience (Siegel et al. 2008).
- **Proximity score** (weight 0.20): inverse-linear score based on distance to nearest source, ranging from 1.0 at distance = 0 to 0.0 at the dispersal limit. Penalises sites close to the dispersal boundary.

Decision: Clustering algorithm

Chosen approach: Union-find at km scale with Gaussian decay kernel

Alternatives considered: DBSCAN; k-nearest-neighbour graph; fixed-radius buffer union; simple distance threshold

- Union-find operates in O(n alpha(n)) time — effectively linear for practical survey sizes (20-10,000 locations). DBSCAN has equivalent complexity but introduces a minPts parameter with no clear biological interpretation for oyster source patches.
- A simple distance threshold (binary: within/outside dispersal range) ignores the exponential decay of larval settlement probability with distance that is consistently documented in field dispersal studies (Robins et al. 2017; Ayata et al. 2010).
- The Gaussian kernel sigma = dispersal_km / 2 places ~95% of larval weight within the full dispersal range, matching the Gaussian plume approximation used in simplified particle tracking (Siegel et al. 2008).

Siegel D.A. et al. (2008): Stochastic nature of larval connectivity among nearshore marine populations. PNAS 105:8974-8979.

Robins P.E. et al. (2017): Empirical and modelled larval dispersal for oysters. J Appl Ecology 54:1699-1710.

Ayata S.D. et al. (2010): Dispersal in a nutrient-limited ocean. Ecology Letters 13:1547-1558.

13.3 External Connectivity Matrix (OpenDrift / FVCOM Route)

When a connectivity_matrix data frame is supplied (columns: source_lat, source_lon, dest_lat, dest_lon, weight [0,1]), the matrix weights are used directly as the connectivity score for matched destination sites. Sites matched to matrix entries take their score from the matrix; unmatched sites fall back to the union-find route. parse_opendrift_connectivity() converts raw OpenDrift settlement output (CSV with release and final lat/lon) into a ready-to-use connectivity matrix by binning particles to a regular grid and computing settlement fractions.

Decision: Particle tracking integration strategy

Chosen approach: Accept pre-computed connectivity matrix; do not run hydrodynamics internally

Alternatives considered: Bundle OpenDrift as a Python dependency; run FVCOM internally; require NetCDF trajectory files

- oystermapR is an R package — bundling a Python hydrodynamic model as a required dependency would violate CRAN policy and make the package uninstallable in standard R environments.
- The connectivity matrix abstraction is flexible: any particle tracking tool (OpenDrift, FVCOM, ROMS-LTRANS, published literature matrices) can supply the same four-column format. The method is not tied to a specific model.
- parse_opendrift_connectivity() handles the most common OpenDrift workflow (release grid + settlement CSV) without requiring users to write their own aggregation code.

Dagestad K.F. et al. (2018): OpenDrift v1.0: a generic framework for trajectory modelling. Geoscientific Model Development 11:1405-1420.

Levin L.A. (2006): Recent progress in understanding larval dispersal: new directions and digressions. Integr Comp Biology 46:282-297.

13.4 Connectivity Classification Scheme

Each site receives one of four connectivity classes based on the composite larval_connectivity_score:

Class	Score range	Ecological interpretation
Highly connected	>= 0.70	Multiple high-quality sources within dispersal range; strong self-replenishment potential
Connected	0.40 - 0.70	Adequate larval supply; suitable for restoration with standard seeding densities
Low connectivity	0.15 - 0.40	Limited larval supply; requires higher initial seeding density; monitor for recruitment failure
Isolated	< 0.15	No reachable source within dispersal range; restoration success depends entirely on initial seeding

Decision: Classification threshold values

Chosen approach: 0.15 / 0.40 / 0.70 breakpoints derived from Gaussian kernel decay at 1/2, 1, and 3/4 sigma

Alternatives considered: Equal-interval quartiles; percentile-based adaptive thresholds; binary connected/isolated

- The 0.15 threshold corresponds to a site receiving larval supply from a single source of moderate quality (0.5) at the dispersal limit — the minimum ecologically plausible contribution before population isolation is functionally complete.
- Binary connected/isolated classifications discard the ecologically meaningful gradient between a site with one nearby source and one with ten clustered sources.
- Adaptive percentile thresholds would shift with every new dataset, making cross-survey comparison impossible — a fixed scale is essential for restoration monitoring over time.

Woolmer A.P. et al. (2009): Ostrea edulis stock assessment and potential for restoration, Firth of Clyde. J Shellfish Research 28:107-116.

14 Remaining Gaps and Future Development

The following items represent the remaining known limitations of oystermapR v0.4.0. Items previously listed as gaps — live data integration, shellfish classification compliance, sediment stability, wave exposure, HAB/predation/anthropogenic risk, Bayesian parameter updating, improved non-edulis species parameterisation, and mechanistic larval dispersal connectivity — have all been implemented. The current outstanding development items are listed below.

14.1 Remaining Gap Summary

Gap / Improvement	Impact	Difficulty	Status
Full hydrodynamic larval dispersal (ROMS/FVCOM coupling)	Medium	Hard	Deferred — hybrid union-find + matrix route implemented in v0.4.0; internal hydrodynamics out of scope for R package
CRAN submission — full test suite pass	High	Moderate	In progress — 136+ tests written; R CMD check --as-cran cleanup in progress
Genetic diversity and sourcing guidance	Low-Med	Hard	Deferred — not a spatial mapping problem
Benthic community facilitation scoring	Low	Hard	Deferred — insufficient field validation data

14.2 CRAN Submission and Test Suite

The test suite has been expanded from 7 tests (test-core.R only) to 136+ tests across 8 test files covering all major modules: core scoring, species parameters, wave/sediment, risk layers, QC, validation, Bayesian updating, variable importance, and larval dispersal. The remaining CRAN check items are: non-ASCII character escaping in R source files, roxygen2 man page generation, and DESCRIPTION/NAMESPACE cleanup. Target: R CMD check --as-cran with zero ERRORs and zero WARNINGs.

14.3 Current Capability Summary (v1.0.0)

Use case	Capability in v1.0.0
Rapid multi-site screening	FULL — all environmental variables, 5 species, QGIS output
Independent restoration planning	FULL — risk layers, larval connectivity, tidal correction, PDF report
Aquaculture permitting support	FULL — shellfish classification, gear feasibility, economic viability, HAB/anthropogenic risk
Published SDM paper	FULL — spatial block CV, permutation importance, Bayesian updating, ODMAP-compliant
Operational monitoring / real-time	PARTIAL — live data via CMEMS/ICES APIs available; requires credentials
Larval connectivity mapping	HYBRID — union-find Gaussian kernel (built-in) + OpenDrift/FVCOM matrix (optional)